Help Desk Lab Documentation

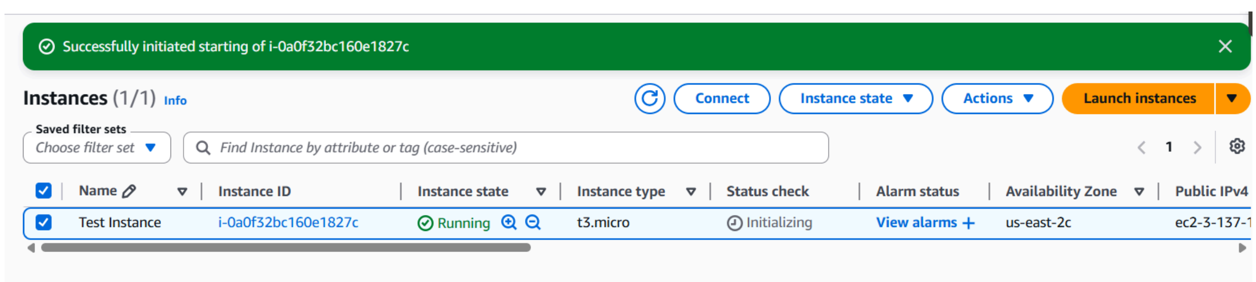
Overview

This lab simulates a real-world IT help desk environment built on Amazon Web Services. A Windows Server 2022 EC2 instance was deployed and configured as a Domain Controller, Active Directory users and groups were created and managed using PowerShell, and all support requests were documented through Zendesk tickets to simulate real help desk workflows. The goal of this lab was to demonstrate hands-on proficiency with the core tools and tasks performed by IT help desk technicians on a daily basis.

Technologies Used

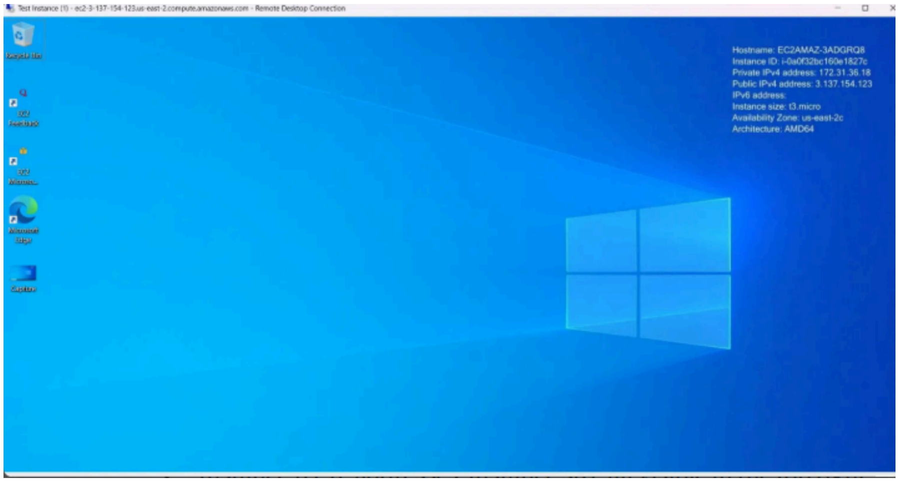
- Amazon Web Services EC2 — Cloud hosting for Windows Server instance
- Windows Server 2022 — Operating system for Domain Controller
- Active Directory Domain Services — User and group management
- PowerShell — Server configuration and AD administration
- Zendesk — Help desk ticketing system
- GitHub — Documentation and project repository

Section 1 — AWS EC2 Instance Setup



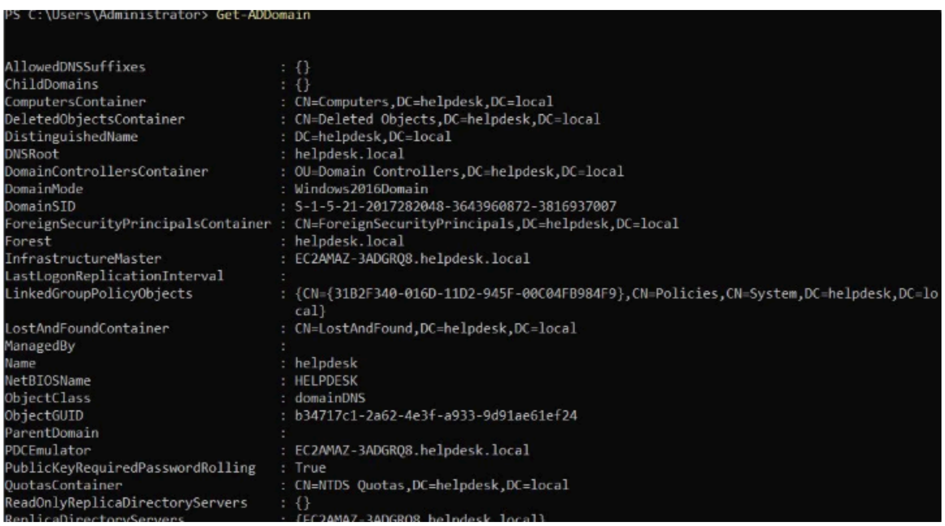
Launched a Windows Server 2022 EC2 instance on AWS using the t3.micro free tier. Configured a security group to allow RDP traffic. Connected to the server remotely using Remote Desktop Protocol with Administrator credentials.

Section 2 — Windows Server Remote Desktop Connection



Connected to the Windows Server 2022 instance via RDP using Administrator credentials. Server hostname confirmed as EC2AMAZ-3ADGRQ8 running on AWS infrastructure.

Section 3 — Domain Controller Configuration



Opened PowerShell as Administrator and ran the following commands:

Command 1: Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools

What this does: Installs the Active Directory software onto the Windows Server. Think of it like installing an app — before you can use Active Directory, the server needs the software installed first. The -IncludeManagementTools part installs the tools you need to manage it.

Command 2: Install-ADDSForest -DomainName "helpdesk.local"

What this does: Promotes the server to a Domain Controller and creates your domain. This is the command that transforms a regular Windows Server into the brain of your network. helpdesk.local is the name of your domain — like a company name for your network.

The server restarted automatically after promotion. Domain confirmed as helpdesk.local.

Section 4 — Active Directory User Account Creation

```
Name              : John Smith
ObjectClass       : user
ObjectGUID        : efc36c91-c415-4178-9f0f-a1650a03883c
SamAccountName    : jsmith
SID               : S-1-5-21-2017282048-3643960872-3816937007-1103
Surname           : Smith
UserPrincipalName : jsmith@helpdesk.loacl

DistinguishedName : CN=Jane Doe,CN=Users,DC=helpdesk,DC=local
Enabled           : True
GivenName         : Jane
Name              : Jane Doe
ObjectClass       : user
ObjectGUID        : 0c8ddd81-f9fc-431b-825d-c61d3dfc3a1d
SamAccountName    : jdoe
SID               : S-1-5-21-2017282048-3643960872-3816937007-1104
Surname           : Doe
UserPrincipalName : jdoe@helpdesk.loacl

DistinguishedName : CN=Bob Johnson,CN=Users,DC=helpdesk,DC=local
Enabled           : True
GivenName         : Bob
Name              : Bob Johnson
ObjectClass       : user
ObjectGUID        : 9967c47c-455d-4714-ab73-c967e4cc1da5
SamAccountName    : bjohnson
SID               : S-1-5-21-2017282048-3643960872-3816937007-1105
Surname           : Johnson
UserPrincipalName : bjohnson@helpdesk.loacl
```

```
PS C:\Users\Administrator> New-ADUser -Name "Jane Doe" -GivenName "Jane" -Surname "Doe" -SamAccountName "jdoe" -UserPrincipalName "jdoe@helpdesk.loacl" -AccountPassword (ConvertTo-SecureString "Password123!" -AsPlainText -Force) -Enabled $true
PS C:\Users\Administrator> New-ADUser -Name "Bob Johnson" -GivenName "Bob" -Surname "Johnson" -SamAccountName "bjohnson" -UserPrincipalName "bjohnson@helpdesk.loacl" -AccountPassword (ConvertTo-SecureString "Password123!" -AsPlainText -Force) -Enabled $true
PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> New-ADUser -Name "John Smith" -GivenName "John" -Surname "Smith" -SamAccountName "jsmith" -UserPrincipalName "jsmith@helpdesk.loacl" -AccountPassword (ConvertTo-SecureString "Password123!" -AsPlainText -Force) -Enabled $true
PS C:\Users\Administrator>
```

Created three user accounts using New-ADUser:

- John Smith — jsmith@helpdesk.local
- Jane Doe — jdoe@helpdesk.local
- Bob Johnson — bjohnson@helpdesk.local

Command: New-ADUser. What this does: Creates a new user account in Active Directory. Every time a new employee joins a company, a help desk technician runs something like this to give them access to the network.

Command: Get-ADUser -Filter * What this does: Pulls up a list of all users in Active Directory. The asterisk means show me everything. Help desk technicians use this to look up user accounts.

Section 5 — User Account Modification

```
PS C:\Users\Administrator> Set-ADUser -Identity "jsmith" -Title "Help Desk Technician" -Department "IT"
PS C:\Users\Administrator>
```

Modified John Smith's account to assign job title as Help Desk Technician and department as IT — simulating a standard onboarding attribute update.

Command: Set-ADUser What this does: Modifies an existing user's information — like updating their job title, department, phone number, or manager. Common during onboarding or when someone changes roles.

Section 6 — Password Reset

```
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "jdoe" -NewPassword (ConvertTo-SecureString "NewPassword456!"
 -AsPlainText -Force) -Reset
PS C:\Users\Administrator>
```

Reset Jane Doe's password, simulating the most common help desk ticket type. A new temporary password was assigned.

Command: Set-ADAccountPassword. What this does: Resets a user's password. The single most common help desk task in existence. Someone gets locked out, you run this command, they're back in.

Section 7 — Account Disable (Employee Offboarding)

```
PS C:\Users\Administrator> Disable-ADAccount -Identity "bjohnson"
PS C:\Users\Administrator>
```

Disabled Bob Johnson's account, simulating an employee offboarding procedure to immediately revoke system access upon departure.

Command: Disable-ADAccount. What this does: Disables a user account without deleting it. When an employee leaves a company, you don't immediately delete their account — you disable it first so their data is preserved, but they can't log in.

Section 8 — Security Group Management

```
PS C:\Users\Administrator> Remove-ADGroupMember -Identity "IT Department" -Members "jsmith" -Confirm:$false
PS C:\Users\Administrator> Get-ADGroupMember -Identity "IT Department"
PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> Add-ADGroupMember -Identity "IT Department" -Members "jsmith"
PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> New-ADGroup -Name "IT Department" -GroupScope Global -GroupCategory Security
PS C:\Users\Administrator>
```

Created an IT Department security group, added and removed John Smith, and verified membership.

Command: New-ADGroup. What this does: Creates a security group. Groups let you manage permissions for multiple users at once — instead of giving 50 people access to a folder one by one, you add them all to a group and give the group access.

Command: Add-ADGroupMember. What this does: Adds a user to a group. When someone joins a department, they get added to that department's security group and automatically inherit all the access that group has.

Command: Remove-ADGroupMember. What this does: Removes a user from a group. When someone changes departments or leaves, you remove them from groups to revoke access.

Command: Get-ADGroupMember. What this does: Shows you who is in a group. Used to verify that a user was successfully added or removed.

Section 9 — User Account Deletion

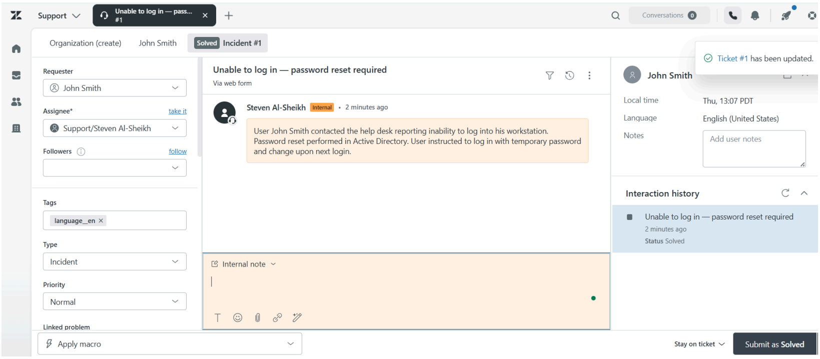
```
PS C:\Users\Administrator> Remove-ADUser -Identity "bjohnson" -Confirm:$false
PS C:\Users\Administrator>
```

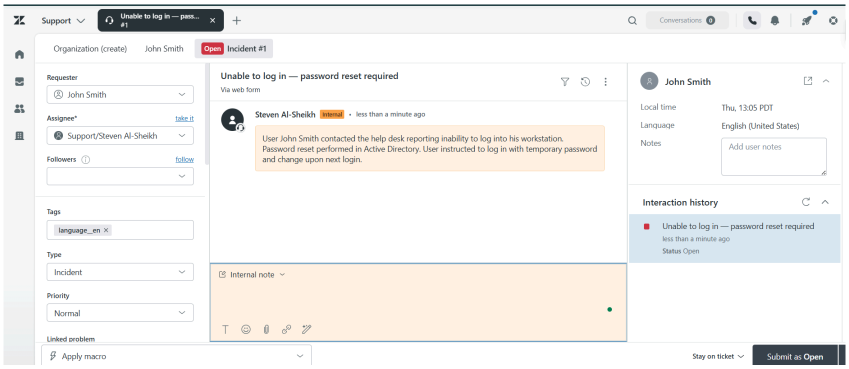
Permanently deleted Bob Johnson's account after offboarding was complete.

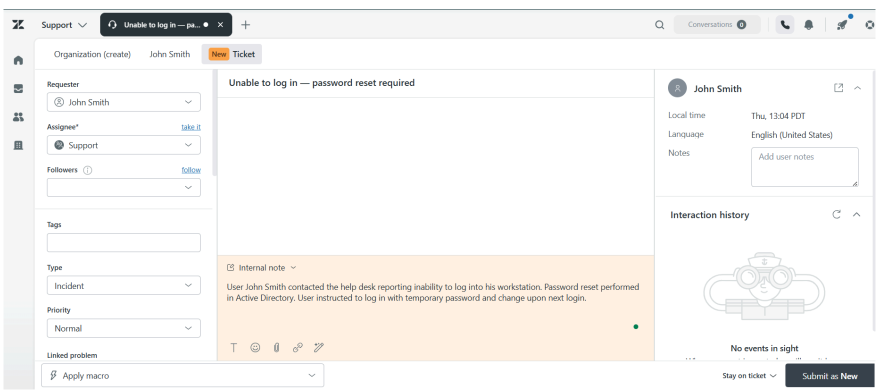
Command: Remove-ADUser. What this does: Permanently deletes a user account. Usually done weeks or months after disabling, once IT confirms the account is no longer needed.

Section 10 — Zendesk Help Desk Tickets

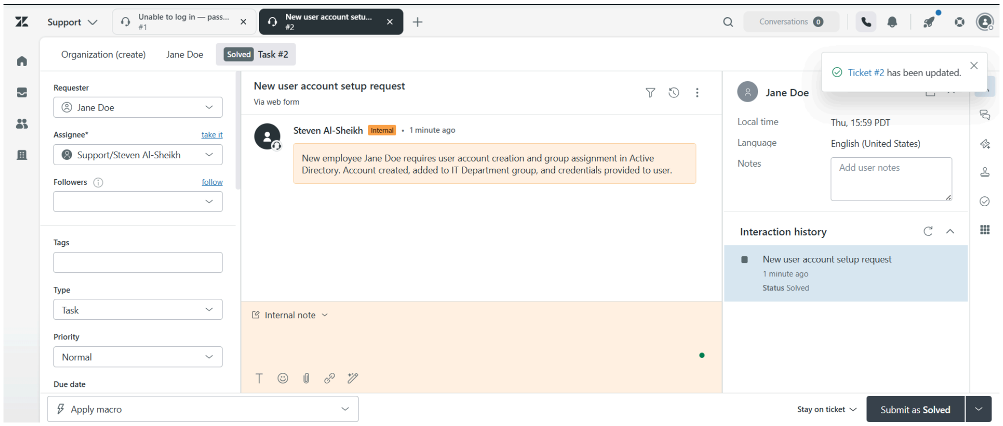
Three support tickets were created and resolved in Zendesk to document the Active Directory tasks performed above.

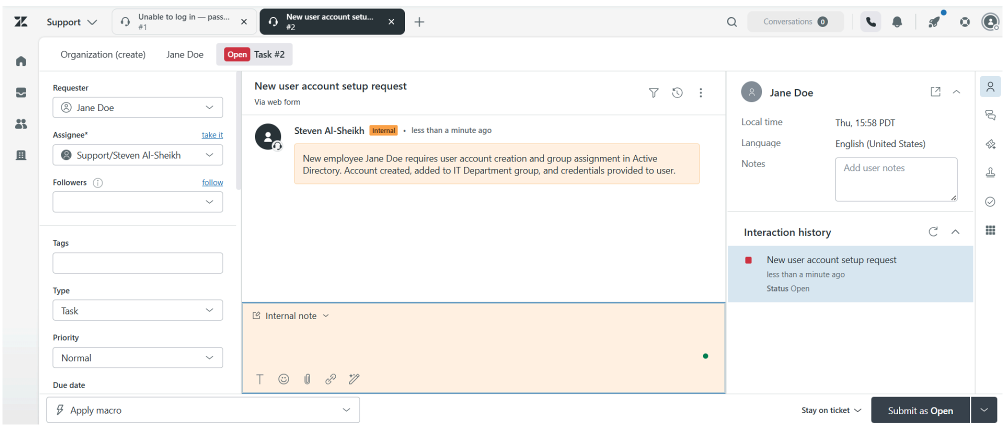




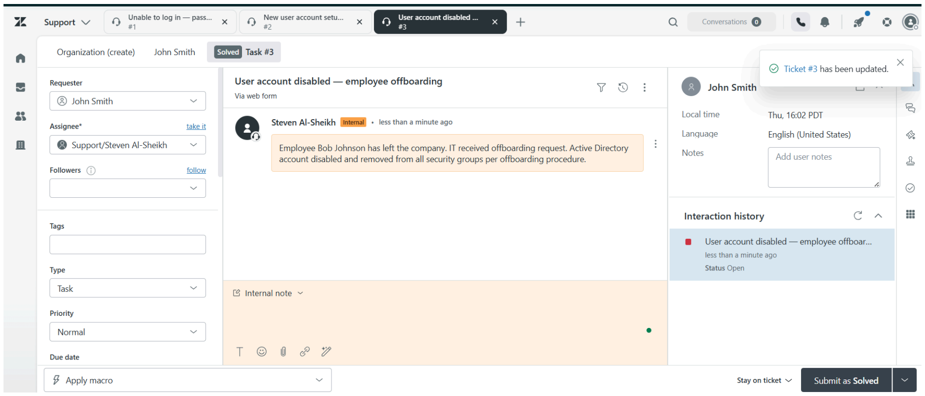


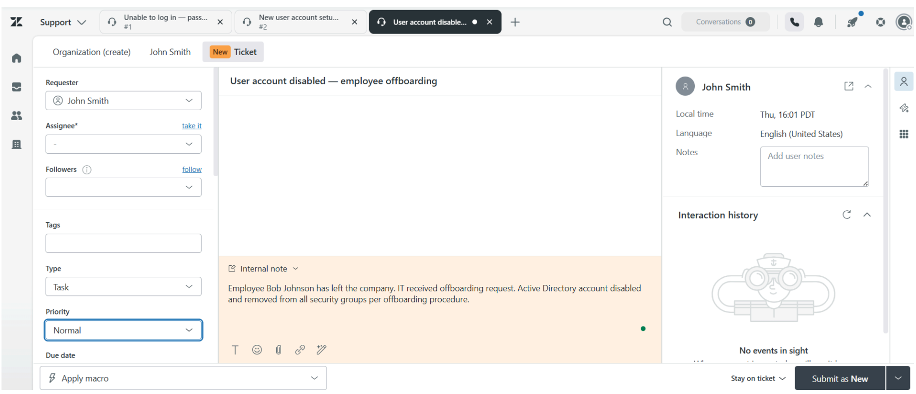
Ticket #1 — Unable to log in — password reset required Requester: John Smith | Type: Incident | Priority: Normal Issue: User John Smith contacted the help desk, reporting inability to log into his workstation. Resolution: Password reset in Active Directory using Set-ADAccountPassword. User instructed to change password on next login. Ticket closed.





Ticket #2 — New user account setup request Requester: Jane Doe | Type: Task | Priority: Normal Issue: New employee Jane Doe requires user account creation and group assignment in Active Directory. Resolution: Account created using New-ADUser and added to the IT Department security group. Credentials provided. Ticket closed.





Ticket #3 — User account disabled — employee offboarding Requester: John Smith | Type: Task | Priority: Normal Issue: Employee Bob Johnson has departed the company. IT received an offboarding request. Resolution: Account disabled using Disable-ADAccount and removed from all security groups. Ticket closed.
